

PUMP A2 / A2B TRIP – TROUBLESHOOTING CHECKLIST

Approved troubleshooting guide for crude charge booster pump A2 (01-A2A) and standby A2B. Revision 2, effective 2024-09-01. Electrical isolation must be done per SOP-EL-014 (current revision) before any hands-on work. Approved by I. Shaikh (Head - Mechanical) and M. Patil (Head - Electrical).

A. Immediate actions (first 15 minutes)

- Confirm standby pump A2B has auto/manually started and crude charge flow (FIC-101) is stable.
- Note trip indication at MCC-2-F07: O/L flag, MCCB trip, or process trip (DCS).
- Check DCS trend for 30 min before trip: motor current, suction pressure PT-2045, discharge flow, vibration.
- Request electrical isolation per SOP-EL-014 before opening terminal box or coupling guard.

B. Mechanical seal leak

- Inspect seal gland for leakage / vapour; check Plan 11 flush line orifice for choking.
- Seal failures on A2: 3 in last 4 years (see work order history). Check for dry-running evidence.
- If leak confirmed, replace seal cartridge; do not restart with visible leakage.

C. Motor overload trip

- Measure IR of M-A2 winding (≥ 1 MOhm at 500 V; typical > 100 MOhm).
- Compare O/L relay setting at MCC-2-F07 with SLD-MCC-2 current revision; check phase currents for imbalance $> 5\%$.
- Check pump for mechanical binding: rotate shaft by hand after isolation.
- Operation beyond rated flow (180 m³/h) increases power demand – verify FIC-101 flow at trip.

D. Low suction pressure / NPSH

- NPSH_r is 3.2 m; check suction strainer dP and suction valve XV-2041 fully open.
- Check V-101 desalter level and crude temperature (vapour pressure).

E. High vibration

- Alarm 4.5 mm/s, trip 7.1 mm/s (RMS). Record DE/NDE readings; check alignment and coupling.
- Check bearing temperature (alarm 85 degC).

F. Restart

- Restart only after cause identified and PTW returned; follow restart steps in SOP-EL-014.
- Log findings in a work order (CM/BD) in the CMMS.

Symptom	Likely cause	Check	Owner
Trip with O/L flag	Overload / binding / low setting	IR, O/L setting, currents	Electrical
Trip + seal pot low	Seal failure	Seal gland, Plan 11 flush	Mechanical
Trip + low PT-2045	Low NPSH / cavitation	Strainer dP, XV-2041	Operations
Vibration trip	Misalignment / bearing	Vibration spectrum	Mechanical